

canyons, dense foliage, and blocked skies reduce accuracy. Multi-constellation and multi-band receivers improve robustness, but RTK and PPP perform best in open-sky conditions. Defining the operating environment early helps guide antenna placement, receiver design, and correction strategy.

Hardware capabilities also impose practical limits. Higher accuracy typically requires greater processing power, increased energy consumption, and higher system costs. Small or battery-powered devices must be evaluated carefully if continuous RTK or PPP operation is required.

Connectivity options include cellular networks (LTE/5G), satellite links, or local base stations, depending on the operating environment. Cellular networks work well in populated areas. Satellite connectivity offers global coverage at a higher cost. Local base stations are well suited to controlled industrial sites.

Antenna and RF design are equally critical. Even the best chipset cannot deliver accurate results if antenna placement is poor, polarisation is incorrect, or RF interference is present. Careful antenna positioning and appropriate filtering often improve performance more effectively than upgrading hardware. Selecting a GNSS architecture therefore requires aligning accuracy requirements, environmental conditions, hardware limitations, connectivity, and RF design within a single integrated solution.

Solution C. Antenna, RF, and PCB design: Where accuracy is won or lost. GNSS accuracy depends on more than the chipset. The antenna, RF front-end, and PCB design are just as important, and in many cases even more influential. A high-end receiver cannot deliver precise results if signal quality is degraded



by a weak antenna or inadequate RF filtering. The antenna is therefore a critical component of the GNSS system.

High-performance GNSS requires clean multi-band signals (L1, L2, L5, and L-Band) from constellations such as the Global Positioning System, GLONASS, Galileo, and BeiDou.

Multi-band support accelerates satellite acquisition, reduces ionospheric errors, and improves signal stability in challenging environments. This capability is essential for achieving centimetre-level accuracy. Urban areas, industrial sites, and nearby electronic equipment generate RF noise, making proper filtering and shielding essential.

Multipath signals reflected from buildings, water, or terrain can distort timing and position calculations. A well-designed antenna helps minimise these effects and improves overall reliability. Companies such

as Taoglas offer multi-band, multi-constellation antennas designed for environments such as urban canyons and dense foliage. These antennas support RTK and PPP solutions by delivering clean signals for both real-time and post-processed corrections.

For engineers, investing in a high-quality antenna and well-designed RF architecture often improves real-world positioning accuracy more than simply upgrading the GNSS chipset.

Solution D. Sensor fusion and system-level design for real-world reliability. Even advanced RTK or PPP solutions cannot eliminate real-world challenges. In urban canyons, dense forests, tunnels, or industrial environments, signals may be blocked, reflected, or interrupted. As a result, centimetre-level accuracy can quickly

degrade.

This degradation is not always caused by algorithms or correction services. In many cases, environmental conditions interfere directly with the signal. True precision is therefore not defined only by theoretical performance but by how reliably a system operates in real-world conditions.

Engineers address GNSS limitations by integrating it with other sensing technologies. Each sensor type has limitations. GNSS struggles in obstructed environments. Inertial measurement units (IMUs) accumulate drift over time. Cameras depend on lighting conditions. Lidar sensors consume significant power.

Sensor fusion enables each sensing modality to compensate for the limitations of the others.

IMUs track motion when GNSS signals are temporarily unavailable. Odometry measures wheel or motor movement. Lidar generates